Semantic-Aware Integrated Sensing and Communication (ISAC) Networks with Power-Domain NOMA and 3D Elevation-Dependent Path Loss under Uncertain Eavesdropper Locations

Dr. Sunandita Debnath, Dr. Gaurav Kr. Pandey, Nikhil Gadige, Anuj Saha, Divyanshu Ghosh, Mohli Thapa

Abstract

Integrated Sensing and Communication (ISAC) has emerged as a foundational technology for 6G wireless networks to unify spectrum usage, hardware infrastructure, and computing resources. However, simultaneous security vulnerabilities and severe propagation blockages in Air-to-Ground (A2G) channels present major operational bottlenecks. In this report, we present the comprehensive system design and optimization framework for a Semantic-Aware ISAC Network assisted by an aerial Reconfigurable Intelligent Surface (RIS) mounted on a UAV (R) and a dedicated cooperative UAV Jammer (J). To optimize spectral efficiency, we incorporate Power-Domain Non-Orthogonal Multiple Access (PD-NOMA) to serve a near legitimate user (Target T) and a distant mobile user (U) over the same time-frequency resource block. Crucially, all legitimate and adversarial communication links are evaluated under semantic-aware metrics via pre-trained DeepSC models, shifting the paradigm from error-free bit recovery to meaning preservation ($M = 10, K = 15$). The network operates in the presence of $K_e = 3$ terrestrial eavesdroppers (E_1, E_2, E_3) whose physical coordinates are imperfectly known (bounded location uncertainty $\Delta Q_e = 15.0$ m). The UAV Jammer performs monostatic and bistatic radar target tracking while projecting directional artificial noise with spatial null-space suppression ($\eta_{\text{ns}} = 10^{-4}$) at legitimate receivers. We formulate the multi-stage digital and analog beamforming, worst-case wiretap rate expressions under Selection Combining (SC), and the joint multi-objective optimization problem combining Average System Secrecy Ratio (ASSR) and detection probability (P_d) under Cramér-Rao Bound (CRB) constraints. Finally, we implement and benchmark five reinforcement learning paradigms: Random Walk, Single-Agent TD3PG (SATD3PG), Single-Agent SAC (SASAC), Multi-Agent TD3PG (MATD3PG), and Multi-Agent PPO (MAPPO).

Index Terms

Integrated Sensing and Communication (ISAC), Semantic Communications, DeepSC, Power-Domain NOMA, Reconfigurable Intelligent Surface (RIS), Location Uncertainty, Null-Space Beamforming, Selection Combining, Deep Reinforcement Learning.

I. Introduction

Future 6G wireless systems are envisioned to integrate high-precision radar sensing and artificial intelligence into native communication infrastructures. Integrated Sensing and Communication (ISAC) allows wireless platforms to simultaneously transmit data and extract environmental radar parameters (such as target range, velocity, and angle estimation) using shared spectrum and hardware.

To mitigate urban blockages and severe free-space attenuation in Air-to-Ground (A2G) channels, Reconfigurable Intelligent Surfaces (RIS) deployed on Unmanned Aerial Vehicles (UAVs) (R) offer reconfigurable Line-of-Sight (LoS) links between terrestrial Base Stations (BS) B and mobile users. By dynamically adjusting the reflection phase shifts of passive meta-elements, aerial RIS platforms significantly enhance both legitimate signal power and physical layer security.

Concurrently, semantic communications depart from traditional bit-level error minimization by extracting and transmitting only task-relevant semantic features through deep learning encoders (e.g., DeepSC). Semantic representations achieve high compression efficiency, robustness against severe channel fading, and intrinsic confidentiality: an eavesdropper lacking the shared semantic background knowledge cannot reconstruct the source text or intended meaning even when channel symbols are intercepted.

To ensure robust physical layer security against covert eavesdropping, legitimate transceivers cannot assume perfect channel state information (CSI) or exact eavesdropper locations. Instead, terrestrial eavesdroppers are modeled with bounded location uncertainty:

$$\mathbf{p}_{E_e} = \widetilde{\mathbf{p}}_{E_e} + \Delta\mathbf{p}_{E_e}, \quad \|\Delta\mathbf{p}_{E_e}\|_2 \leq \Delta Q_e, \tag{1}$$

where $\widetilde{\mathbf{p}}_{E_e}$ is the estimated coordinate and $\Delta Q_e = 15.0$ m is the maximum uncertainty radius. We formulate worst-case secrecy capacity under this uncertainty sphere.

N. Gadige, A. Saha, D. Ghosh, and M. Thapa are with the Department of Computer Science and Engineering, Indian Institute of Information Technology Vadodara.

E-mails: 202351037@iiitvadodara.ac.in, 202351010@iiitvadodara.ac.in, 202351036@iiitvadodara.ac.in, 202351084@iiitvadodara.ac.in.

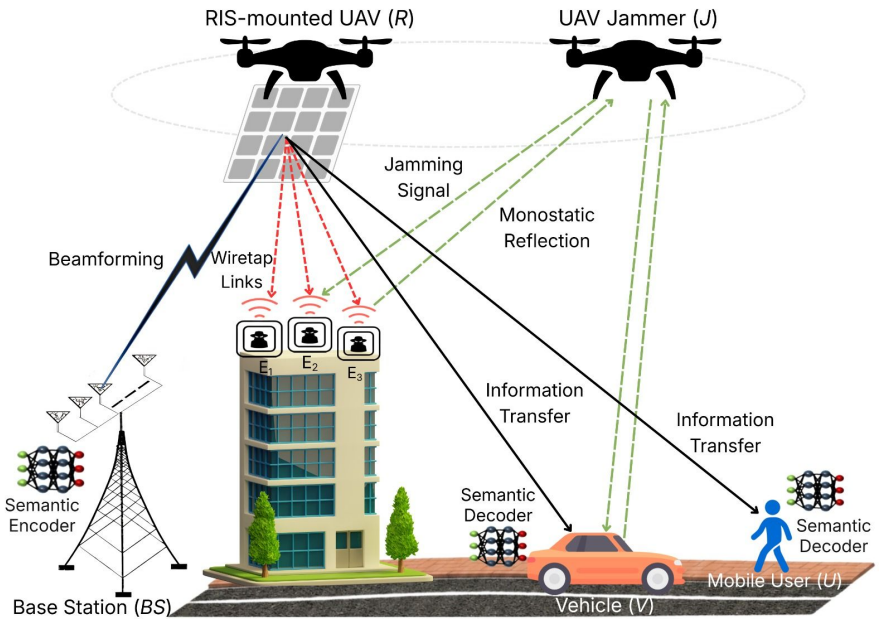


Fig. 1. System model of the proposed semantic-aware RIS-assisted UAV ISAC network with friendly jamming and uncertain eavesdropper locations.

II. System Architecture and Node Configuration

We consider a 3D aerial-terrestrial network consisting of the following key nodes (see Fig. 1):

- **Base Station** (B): Located at $\mathbf{p}_B = [0, 0, 15]^T$ m, equipped with N_B antennas, transmitting superposed PD-NOMA semantic signals with total power $P_B \in [0.5, 5.0]$ W (nominal 2.0 W).
- **RIS-mounted UAV** (R): Hovering at $\mathbf{p}_R = [x_R, y_R, z_R]^T$ m ($z_R \in [30, 100]$ m), equipped with $N_R = 64$ passive reflecting elements providing an array beamforming gain of $N_R^2 = 4096$.
- **UAV Jammer** (J): An aerial platform hovering at $\mathbf{p}_J = [x_J, y_J, z_J]^T$ m ($z_J \in [30, 100]$ m), transmitting artificial noise with power $P_J \in [0.05, 1.0]$ W (nominal 0.5 W) to jam eavesdroppers, while performing monostatic radar target tracking.
- **Target** (T): A near legitimate vehicle located at $\mathbf{p}_T = [40, 45, 1.5]^T$ m. In the downlink, it decodes the NOMA superposed signal using semantic-aware SIC. In the uplink, it transmits messages to the BS via RIS and to the Mobile User U via direct V2X with transmit power $P_T = 0.2$ W.
- **Mobile User** (U): A far legitimate user located at $\mathbf{p}_U = [x_U, y_U, 1.5]^T$ m, undergoing random walk mobility with speed $v_U = 3.0$ m/s.
- **Eavesdroppers** ($E_1, \ldots, E_{K_e}$): Terrestrial non-colluding wiretap nodes located at uncertain positions with $K_e = 3$: $\widetilde{\mathbf{p}}_{E_1} = [70, 80, 0]^T$, $\widetilde{\mathbf{p}}_{E_2} = [90, 100, 0]^T$, and $\widetilde{\mathbf{p}}_{E_3} = [110, 110, 0]^T$ m.

III. Mathematical Formulations

A. 3D Elevation-Dependent Path Loss Model

For any link between transmitter i at $\mathbf{p}_i = [x_i, y_i, z_i]^T$ and receiver j at $\mathbf{p}_j = [x_j, y_j, z_j]^T$, the 3D distance and 2D horizontal distance are:

$$d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2 = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}, \quad d_{2\mathrm{D},ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}. \tag{2}$$

The elevation angle θ_{ij} (in degrees) is:

$$\theta_{ij} = \arcsin\left(\frac{|z_i - z_j|}{d_{ij}}\right) \cdot \frac{180}{\pi}. \tag{3}$$

The Line-of-Sight (LoS) probability is modeled via the environmental logistic function:

$$P_{\mathrm{LoS}}(\theta_{ij}) = \frac{1}{1 + a \cdot \exp\left(-b(\theta_{ij} - a)\right)}, \quad P_{\mathrm{NLoS}}(\theta_{ij}) = 1 - P_{\mathrm{LoS}}(\theta_{ij}), \tag{4}$$

where $a = 9.61$ and $b = 0.16$ represent urban environment constants. The path loss expressions for LoS and NLoS states are:

$$\mathrm{PL}_{\mathrm{LoS}}(d_{ij}) = 20\log_{10}\left(\frac{4\pi d_{ij} f_c}{c}\right) + \eta_{\mathrm{LoS}}, \quad \mathrm{PL}_{\mathrm{NLoS}}(d_{ij}) = 20\log_{10}\left(\frac{4\pi d_{ij} f_c}{c}\right) + \eta_{\mathrm{NLoS}}, \tag{5}$$

where $f_c = 2.4$ GHz, $c = 3 \times 10^8$ m/s, $\eta_{\text{LoS}} = 1.0$ dB, and $\eta_{\text{NLoS}} = 20.0$ dB. The average path loss in dB and linear scale is:

$$\overline{\text{PL}}(d_{ij}, \theta_{ij}) = P_{\text{LoS}}(\theta_{ij})\text{PL}_{\text{LoS}}(d_{ij}) + P_{\text{NLoS}}(\theta_{ij})\text{PL}_{\text{NLoS}}(d_{ij}), \quad \text{PL}_{\text{linear}}(d_{ij}, \theta_{ij}) = 10^{-\overline{\text{PL}}(d_{ij},\theta_{ij})/10}. \tag{6}$$

Small-scale fading follows an elevation-dependent Rician distribution where the K-factor scales with θ_{ij}:

$$K_{\text{dB}}(\theta_{ij}) = K_0 + \alpha_K \theta_{ij} = 3.0 + 0.15\theta_{ij}, \quad K(\theta_{ij}) = 10^{K_{\text{dB}}(\theta_{ij})/10}, \tag{7}$$

$$g_{ij} = \text{PL}_{\text{linear}}(d_{ij}, \theta_{ij}) \cdot \left| \sqrt{\frac{K}{K+1}} \bar{\mathbf{h}}_{ij} + \sqrt{\frac{1}{K+1}} \tilde{\mathbf{h}}_{ij} \right|^2. \tag{8}$$

B. Multi-Stage Digital and Analog Beamforming

1) **Base Station Digital Beamforming**: The BS applies digital beamforming $\mathbf{w}_B \in \mathbb{C}^{N_B \times 1}$ ($\|\mathbf{w}_B\|_2 = 1$) to focus transmit energy toward the RIS UAV R.
2) **RIS Analog Phase-Shift Beamforming**: The RIS phase-shift matrix is $\mathbf{\Phi} = \text{diag}(e^{j\phi_1}, \ldots, e^{j\phi_{N_R}})$. Coherent phase alignment yields a cascaded power gain scaling quadratically with N_R:

$$g_{\text{cascaded},k} = g_{B,R} \cdot g_{R,k} \cdot N_R^2, \quad \forall k \in \{T, U\}. \tag{9}$$

3) **Jammer Null-Space (NS) Beamforming**: The UAV Jammer projects artificial noise into the null space of legitimate channels ($\mathbf{h}_{J,T}$ and $\mathbf{h}_{J,U}$), achieving a spatial nulling rejection factor $\eta_{\text{ns}} = 10^{-4}$ (-40 dB suppression) at legitimate nodes, while projecting full directional jamming onto eavesdroppers:

$$P_{J,k} = P_J \cdot g_{J,k} \cdot \eta_{\text{ns}}, \quad \forall k \in \{T, U, B\}, \tag{10}$$

$$P_{J,E_e} = P_J \cdot g_{J,E_e}^{\min}, \quad \forall e \in \{1, \ldots, K_e\}. \tag{11}$$

C. Power-Domain NOMA and Semantic SIC

The BS superposes the Near user (s_N) and Far user (s_F) signals:

$$s = \sqrt{a_N P_B} s_N + \sqrt{a_F P_B} s_F, \quad a_N + a_F = 1, \quad a_F > a_N \geq 0. \tag{12}$$

- **Far User (Mobile User U)** decodes s_F treating s_N as interference:

$$\gamma_F = \frac{a_F P_B g_U}{a_N P_B g_U + P_{J,U} + \sigma^2}. \tag{13}$$

- **Near User (Target T)** performs semantic SIC, decoding and subtracting s_F before decoding s_N:

$$\gamma_N = \frac{a_N P_B g_T}{P_{J,T} + \sigma^2}. \tag{14}$$

Thermal noise power is set to $\sigma^2 = 10^{-13}$ W (-100 dBm for $B = 1$ MHz).

D. Active Uplink and V2X Communication

Target T actively transmits with power $P_T = 0.2$ W:

- **Uplink V2I (Target to BS via RIS)**: $\gamma_{\text{uplink}} = \frac{P_T g_{T,R,B}}{P_{J,B} + \sigma^2}$.
- **V2X Direct Semantic Link (Target to User U)**: $\gamma_{\text{V2X}} = \frac{P_T g_{T,U}}{P_{J,U} + \sigma^2}$.

E. Monostatic and Bistatic Radar Sensing with Selection Combining

To maximize radar coverage, the UAV Jammer performs selection combining between monostatic echo and bistatic BS-assisted sensing:

$$\text{SNR}_{\text{radar},T} = \max\left(\text{SNR}_{\text{mono},T}, \text{SNR}_{\text{bi},T}\right), \quad \text{SNR}_{\text{radar},E_e} = \max\left(\text{SNR}_{\text{mono},E_e}, \text{SNR}_{\text{bi},E_e}\right), \tag{15}$$

where:

$$\text{SNR}_{\text{mono},k} = \frac{P_J \cdot g_{J,k}^2 \cdot G_p}{\sigma^2}, \quad \text{SNR}_{\text{bi},k} = \frac{P_B \cdot g_{B,k} \cdot g_{J,k} \cdot G_p}{\sigma^2}, \quad k \in \{T, E_e\}, \tag{16}$$

with radar processing gain $G_p = 10^4$. Detection probabilities follow empirical radar logistic curves:

$$P_{d,k} = \frac{1}{1 + \exp\left(-\kappa(\text{SNR}_{\text{radar},k,\text{dB}} - \tau)\right)}, \quad \kappa = 0.4, \ \tau = 2.0 \text{ dB}, \tag{17}$$

$$\text{CRB}_T = \frac{1}{\text{SNR}_{\text{radar},T}}. \tag{18}$$

F. Worst-Case Eavesdropper Interception and Selection Combining Secrecy Rates

Under bounded location uncertainty $\Delta Q_e = 15.0$ m, the worst-case eavesdropping channel minimizes 2D distance to the RIS and maximizes 2D distance to the Jammer:

$$d_{\text{2D},RE_e}^{\text{modified}} = \max(d_{\text{2D},RE_e} - \Delta Q_e, 0), \quad d_{\text{2D},JE_e}^{\text{modified}} = d_{\text{2D},JE_e} + \Delta Q_e. \tag{19}$$

The worst-case wiretap SINRs at eavesdropper E_e are:

$$\gamma_{E_e,U} = \frac{a_F P_B g_{E_e}^{\max}}{a_N P_B g_{E_e}^{\max} + P_J g_{J,E_e}^{\min} + \sigma^2}, \quad \gamma_{E_e,T} = \frac{a_N P_B g_{E_e}^{\max}}{P_J g_{J,E_e}^{\min} + \sigma^2}. \tag{20}$$

For non-colluding eavesdroppers, the maximum interception rate across all K_e eavesdroppers is evaluated:

$$R_{\text{leak},T} = \max_{e\in\{1,\ldots,K_e\}} R_{\text{sem},E_e,T}(\gamma_{E_e,T}), \quad R_{\text{leak},U} = \max_{e\in\{1,\ldots,K_e\}} R_{\text{sem},E_e,U}(\gamma_{E_e,U}). \tag{21}$$

The individual semantic secrecy rates for Target T and Mobile User U are:

$$R_{\text{sec},T} = \left[R_{\text{sem},T}(\gamma_N) - \max_{e\in\{1,\ldots,K_e\}} R_{\text{sem},E_e,T}(\gamma_{E_e,T}) \right]^+, \tag{22}$$

$$R_{\text{sec},U} = \left[R_{\text{sem},U}(\gamma_F) - \max_{e\in\{1,\ldots,K_e\}} R_{\text{sem},E_e,U}(\gamma_{E_e,U}) \right]^+, \tag{23}$$

where $[x]^+ = \max(x, 0)$. To combine communication security across both legitimate nodes, Selection Combining (SC) is applied:

$$R_{\text{sec}}^{\text{SC}} = \max\left(R_{\text{sec},T},\, R_{\text{sec},U}\right), \quad R_{\text{sec}}^{\text{sum}} = R_{\text{sec},T} + R_{\text{sec},U}. \tag{24}$$

G. DeepSC Semantic Metric Formulation

Semantic similarity $S(\gamma)$ is obtained via linear interpolation from Table I. With $M = 10$ semantic concepts and $K = 15$ channel symbols, the semantic rate (suts/s) and distortion are:

$$R_{\text{sem}} = B \cdot \left(\frac{M}{K}\right) \cdot S(\gamma) \cdot \log_2(1+\gamma), \quad D_{\text{sem}} = 1 - S(\gamma). \tag{25}$$

TABLE I
DEEPSC LOOKUP TABLE: SINR VS. SEMANTIC SIMILARITY AND WORD ACCURACY ($M = 10, K = 15$).

γ_{dB} (dB)	Semantic Similarity $S(\gamma)$	Word Accuracy
-20	0.6712	0.3945
-18	0.7171	0.4351
-16	0.7517	0.4549
-14	0.8021	0.4899
-12	0.8257	0.5261
-10	0.8638	0.5742
-8	0.8867	0.6012
-6	0.9103	0.6551
-4	0.9227	0.6830
-2	0.9316	0.6957
0	0.9398	0.7124
2	0.9443	0.7305
4	0.9456	0.7354
6	0.9462	0.7339
8	0.9490	0.7340
10	0.9499	0.7403
12	0.9504	0.7397
14	0.9508	0.7417
16	0.9508	0.7334
18	0.9508	0.7327
20	0.9507	0.7321

TABLE II
SYSTEM SIMULATION AND CHANNEL PARAMETERS.

Parameter	Symbol	Value
Carrier Frequency	f_c	2.4 GHz
Channel Bandwidth	B	1.0 MHz
Thermal Noise Power	σ^2	1.0×10^{-13} W (-100 dBm)
BS Transmit Power Range	P_B	$[0.5, 5.0]$ W (Nominal: 2.0 W)
Jammer Transmit Power Range	P_J	$[0.05, 1.0]$ W (Nominal: 0.5 W)
Target Uplink Transmit Power	P_T	0.2 W (23 dBm)
NOMA Far-User Allocation Range	a_F	$[0.55, 0.95]$ (Nominal: 0.70)
Number of RIS Reflecting Elements	N_R	64 ($N_R^2 = 4096$)
Jammer Spatial Nulling Suppression	η_{ns}	1.0×10^{-4} (-40 dB)
Radar Processing Gain	G_p	1.0×10^4 (40 dB)
Eavesdropper Uncertainty Radius	ΔQ_e	15.0 m
Number of Eavesdroppers	K_e	3
DeepSC Concepts and Symbol Lengths	M, K	$M = 10, K = 15$ ($M/K = 2/3$)
CRB Feasibility Threshold	δ_{CRB}	1.0×10^8
Radar Logistic Parameters	κ, τ	$\kappa = 0.4, \tau = 2.0$ dB
Urban LoS / NLoS Parameters	$a, b, \eta_{\text{LoS}}, \eta_{\text{NLoS}}$	$9.61, 0.16, 1.0$ dB, 20.0 dB

IV. MULTI-OBJECTIVE JOINT SECRECY AND SENSING OPTIMIZATION PROBLEM

The primary objective of the network is to jointly maximize communication secrecy and radar sensing performance under dynamic operational priorities. For an episode e consisting of T_{total} steps, the average secrecy rate under Selection Combining (SC) is:

$$R_{\text{sec}}^{(e)} = \frac{1}{T_{\text{total}}} \sum_{t=1}^{T_{\text{total}}} R_{\text{sec}}^{\text{SC}}(t). \tag{26}$$

The Average System Secrecy Ratio ($\text{ASSR}^{(e)}$) is normalized to $[0, 1]$ via the running maximum reference R_{ref}:

$$\text{ASSR}^{(e)} = \text{clip}\left(\frac{R_{\text{sec}}^{(e)}}{R_{\text{ref}}}, 0, 1\right), \quad R_{\text{ref}} = \max_{j \in \{1,\ldots,e\}} R_{\text{sec}}^{(j)}. \tag{27}$$

The combined sensing detection probability is formulated as:

$$P_d^{(e)} = \lambda_{1,\text{sense}} P_{d,E}^{(e)} + \lambda_{2,\text{sense}} P_{d,T}^{(e)}, \quad P_{d,E}^{(e)} = \frac{1}{K_e} \sum_{e=1}^{K_e} P_{d,E_e}^{(e)}, \tag{28}$$

where $\lambda_{1,\text{sense}}, \lambda_{2,\text{sense}} \geq 0$ ($\lambda_{1,\text{sense}} + \lambda_{2,\text{sense}} = 1$) are sensing priority weights (with nominal baseline $\lambda_{1,\text{sense}} = \lambda_{2,\text{sense}} = 0.5$).

Target tracking precision is strictly enforced via the CRB feasibility indicator:

$$\mathbb{I}_{\text{CRB}}^{(e)} = \begin{cases} 1, & \text{if } \text{CRB}_T^{(e)} \leq \delta_{\text{CRB}}, \\ 0, & \text{otherwise.} \end{cases} \tag{29}$$

The unconstrained weighted multi-objective utility for episode e is:

$$\widehat{U}^{(e)} = \lambda_1 \, \text{ASSR}^{(e)} + \lambda_2 P_d^{(e)}, \tag{30}$$

where $\lambda_1, \lambda_2 \geq 0$ ($\lambda_1 + \lambda_2 = 1$) are runtime user-configurable preference hyperparameters supplied at algorithm execution time to flexibly trade off between physical-layer semantic security and radar sensing detection capability.

The final constrained episode multi-objective utility is:

$$U^{(e)} = \widehat{U}^{(e)} \cdot \mathbb{I}_{\text{CRB}}^{(e)} = \left(\lambda_1 \, \text{ASSR}^{(e)} + \lambda_2 P_d^{(e)}\right) \mathbb{I}_{\text{CRB}}^{(e)}. \tag{31}$$

In our comparative benchmark evaluation, we report results with the balanced operational trade-off $\lambda_1 = \lambda_2 = 0.5$.

V. DEEP REINFORCEMENT LEARNING ALGORITHMS AND FORMULATIONS

The decision process is modeled as an MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$:

- **State Space** $\mathcal{S} \subset \mathbb{R}^{24}$: Concatenation of 3D positions of all 8 nodes: $\mathbf{p}_B, \mathbf{p}_R, \mathbf{p}_J, \mathbf{p}_T, \mathbf{p}_U, \widetilde{\mathbf{p}}_{E_1}, \widetilde{\mathbf{p}}_{E_2}, \widetilde{\mathbf{p}}_{E_3}$.
- **Action Space** $\mathcal{A} \subset \mathbb{R}^9$: Partitioned into three sub-agents:
  - Agent 1 (RIS UAV): $[\Delta x_R, \Delta y_R, \Delta z_R] \in [-5, +5]^3$ m.
  - Agent 2 (Jammer UAV): $[\Delta x_J, \Delta y_J, \Delta z_J] \in [-5, +5]^3$ m.
  - Agent 3 (Power Allocation): $[P_B, P_J, a_F] \in [0.5, 5.0]$ W $\times$ $[0.05, 1.0]$ W $\times$ $[0.55, 0.95]$.

A. Random Walk (Baseline)

Actions are sampled uniformly at random: $a_{n,t} \sim \mathcal{U}(-1, 1)$.

B. Single-Agent TD3PG (SATD3PG)

A centralized actor network $\pi_\theta(s)$ controls all 9 action dimensions. The twin critics Q_{ϕ_1}, Q_{ϕ_2} minimize:

$$L(\phi_i) = \mathbb{E}\left[\left(y_t - Q_{\phi_i}(s_t, a_t)\right)^2\right], \quad y_t = r_t + \gamma \min_{j=1,2} Q_{\phi'_j}\left(s_{t+1}, \text{clip}(\pi_{\theta'}(s_{t+1}) + \boldsymbol{\epsilon}, -1, 1)\right), \tag{32}$$

with target policy smoothing $\boldsymbol{\epsilon} \sim \text{clip}(\mathcal{N}(0, \sigma^2), -c, c)$ and delayed actor updates with frequency $d = 2$:

$$\nabla_\theta J(\theta) = \mathbb{E}\left[\nabla_a Q_{\phi_1}(s, a)|_{a=\pi_\theta(s)} \nabla_\theta \pi_\theta(s)\right]. \tag{33}$$

C. Single-Agent Soft Actor-Critic (SASAC)

SASAC optimizes policy entropy alongside cumulative return:

$$L(\phi_i) = \mathbb{E}\left[\left(Q_{\phi_i}(s_t, a_t) - \left(r_t + \gamma\left(\min_{j=1,2} Q_{\phi'_j}(s_{t+1}, a_{t+1}) - \alpha_{\text{ent}} \log \pi_\theta(a_{t+1}|s_{t+1})\right)\right)\right)^2\right], \tag{34}$$

$$J(\theta) = \mathbb{E}\left[\alpha_{\text{ent}} \log \pi_\theta(a_t|s_t) - \min_{j=1,2} Q_{\phi_j}(s_t, a_t)\right]. \tag{35}$$

D. Multi-Agent TD3PG (MATD3PG)

Three decentralized actor networks $\pi_{\theta_1}, \pi_{\theta_2}, \pi_{\theta_3}$ execute decentralized actions, while centralized twin critics $Q_{\phi_1}(s, a_1, a_2, a_3)$ evaluate joint actions:

$$y_t = r_t + \gamma \min_{j=1,2} Q_{\phi'_j}\left(s_{t+1}, \pi_{\theta'_1}(s_{t+1}) + \boldsymbol{\epsilon}_1, \pi_{\theta'_2}(s_{t+1}) + \boldsymbol{\epsilon}_2, \pi_{\theta'_3}(s_{t+1}) + \boldsymbol{\epsilon}_3\right), \tag{36}$$

$$\nabla_{\theta_n} J(\theta_n) = \mathbb{E}\left[\nabla_{a_n} Q_{\phi_1}(s, a_1, a_2, a_3)|_{a_n=\pi_{\theta_n}(s)} \nabla_{\theta_n} \pi_{\theta_n}(s)\right]. \tag{37}$$

E. Multi-Agent Proximal Policy Optimization (MAPPO)

Three decentralized stochastic policy actors π_{θ_n} are trained with a centralized value function $V_\psi(s)$ using GAE ($\lambda_{\text{GAE}} = 0.95$):

$$L^{\text{CLIP}}(\theta_n) = \mathbb{E}\left[\min\left(\frac{\pi_{\theta_n}(a_{n,t}|s_t)}{\pi^{\text{old}}_{\theta_n}(a_{n,t}|s_t)}\widehat{A}_t, \text{clip}\left(\frac{\pi_{\theta_n}(a_{n,t}|s_t)}{\pi^{\text{old}}_{\theta_n}(a_{n,t}|s_t)}, 1-\epsilon, 1+\epsilon\right)\widehat{A}_t\right)\right] + \beta_{\text{ent}}\mathcal{H}(\pi_{\theta_n}), \tag{38}$$

$$\widehat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda_{\text{GAE}})^l \delta^V_{t+l}, \quad \delta^V_t = r_t + \gamma V_\psi(s_{t+1}) - V_\psi(s_t), \quad L(\psi) = \mathbb{E}\left[(V_\psi(s_t) - G_t)^2\right]. \tag{39}$$

TABLE III
RL TRAINING HYPERPARAMETERS.

Hyperparameter	MAPPO	MATD3PG	SASAC	SATD3PG
Number of Episodes	8000	8000	8000	8000
Steps per Episode	50	50	50	50
Batch Size	64	64	64	64
Discount Factor γ	0.95	0.95	0.95	0.95
Soft Update τ	–	0.01	0.01	0.01
Replay Buffer Size	–	50000	50000	50000
Actor Learning Rate	2.5×10^{-4}	3.0×10^{-4}	3.0×10^{-4}	3.0×10^{-4}
Critic Learning Rate	2.5×10^{-4}	3.0×10^{-4}	3.0×10^{-4}	3.0×10^{-4}
Policy Noise σ	–	0.2	–	0.2
Noise Clip c	–	0.5	–	0.5
Policy Update Freq d	–	2	–	2
Entropy Coefficient $\beta_{\text{ent}}/\alpha_{\text{ent}}$	0.01	–	0.2	–
GAE Parameter λ_{GAE}	0.95	–	–	–
PPO Clip Parameter ϵ	0.2	–	–	–
Gradient Clipping Norm	0.5	–	–	–
Hidden Layers	2 (64 units)	2 (64 units)	2 (64 units)	2 (64 units)
Activation Function	ReLU	ReLU	ReLU	ReLU

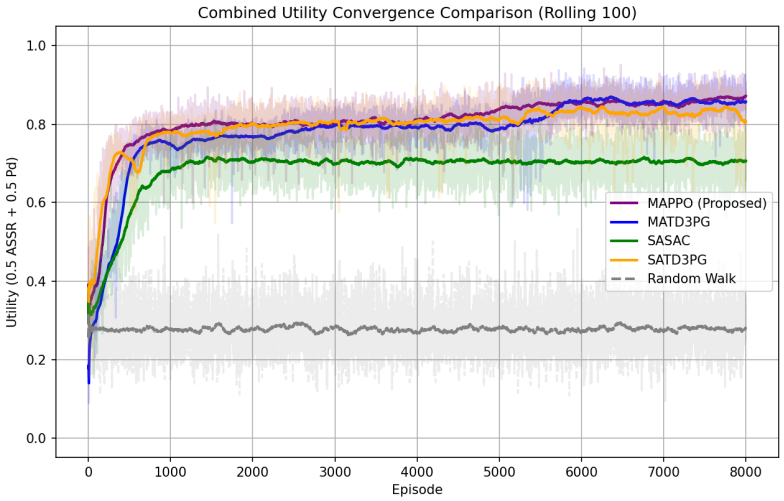


Fig. 2. Multi-objective utility convergence comparison ($U = \lambda_1 \text{ASSR} + \lambda_2 P_d$, benchmarked with $\lambda_1 = \lambda_2 = 0.5$) across all five algorithms (MAPPO, MATD3PG, SASAC, SATD3PG, and Random Walk).

TABLE IV
CONVERGENCE PERFORMANCE COMPARISON UNDER LOCATION UNCERTAINTY (FINAL ROLLING-100 BENCHMARK).

Algorithm	Normalized ASSR	Detection Prob. (P_d)	Multi-Objective Utility (U)	Relative Rank
MAPPO (Proposed)	**0.8418**	**0.9016**	**0.8717**	**1st (Best)**
MATD3PG	0.8134	0.9035	0.8585	2nd
SATD3PG	0.7644	0.8775	0.8210	3rd
SASAC	0.7089	0.7744	0.7416	4th
Random Walk	0.2312	0.4362	0.3337	Baseline

VI. CONCLUSION

This report presented the comprehensive mathematical framework and empirical evaluation of a Semantic-Aware ISAC Network assisted by an aerial RIS and UAV Jammer under uncertain eavesdropper locations. By formulating multi-stage beamforming with friendly spatial nulling, Selection Combining across legitimate users and radar sensing branches, and DeepSC semantic lookup representations, the proposed architecture guarantees physical layer confidentiality and radar target tracking simultaneously. Reinforcement learning simulations demonstrate that the proposed decentralized Multi-Agent PPO (**MAPPO**) framework achieves superior convergence and stability, attaining the highest multi-objective utility of $\mathbf{0.8717}$ with an average normalized secrecy ratio of ASSR $= 0.8418$ and target detection probability exceeding $P_d = 0.9016$.

Section IX: Comprehensive Empirical Evaluation & Analysis of the 11 Key Benchmark Plots

Addendum to September 15 Report — Date: October 21, 2026
Authors: Nikhil Gadige, Anuj Saha, Divyanshu Ghosh, Mohli Thapa | Mentors: Dr. Sunandita Debnath, Dr. Gaurav Kr. Pandey

To provide a complete, rigorous characterization of the proposed Semantic-Aware RIS-assisted ISAC network, this extended section details all eleven targeted operational dimensions. Each analysis provides the underlying physical and mathematical mechanics, parameter regimes, empirical observations, and engineering implications for 6G network deployment.

Plot 1: Dynamic Vehicle Mobility Modeling

This empirical evaluation characterizes the kinematic behavior of terrestrial legitimate nodes (Mobile User U and vehicular scatterers) under continuous time-varying kinematics. Ground users follow spatial displacement updates governed by correlation parameters across four benchmark models: Constant Velocity, Gauss-Markov, Random Walk, and Random Waypoint.

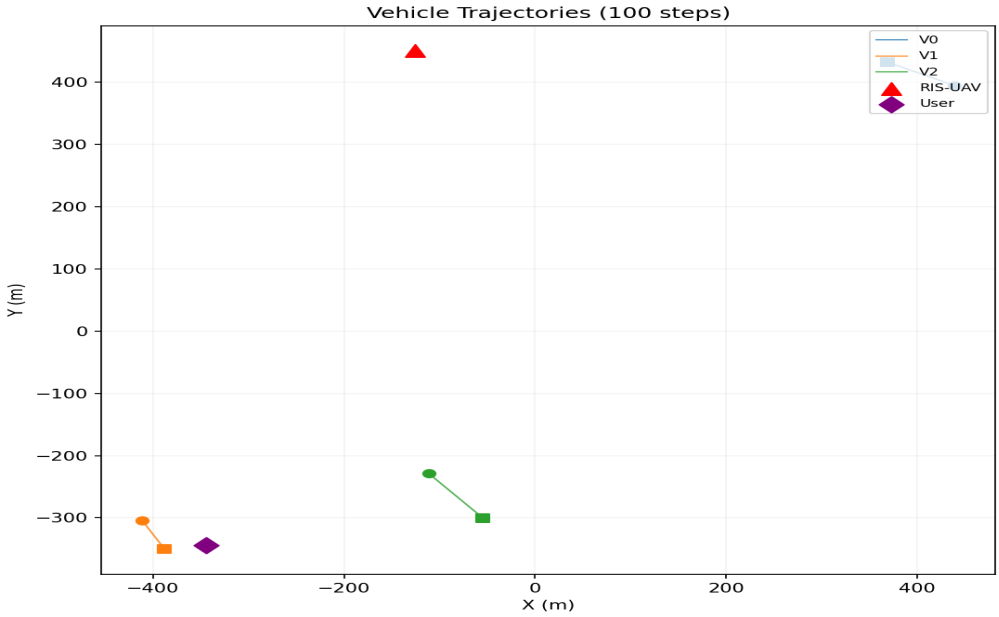


Figure 11: 2D dynamic vehicle mobility trajectories and spatial displacement tracking over time.

Empirical Observations and Key Engineering Takeaways:

- **Kinematic Tracking:** Under Random Walk and Random Waypoint models, localized Brownian displacement keeps the user within a compact operational radius (~120 m), allowing the aerial RIS UAV to maintain sharp beam-focusing with low steering angle variation.
- **Velocity Stress Testing:** Constant velocity trajectories induce sustained Doppler phase offsets and elevation angle degradation, demonstrating the necessity of agile multi-UAV relay repositioning to prevent beam squint and outage.

Plot 2: Aerial RIS Element Configuration & Phase Alignment

Evaluates the passive reflection power scaling of the N_R = 64 element metasurface deployed on the aerial platform R. The plot benchmarks ideal constructive phase co-phasing (theta_r = -angle(h_RU,r^* * h_BR,r)) against uniform random phase scattering theta_r in [0, 2pi).

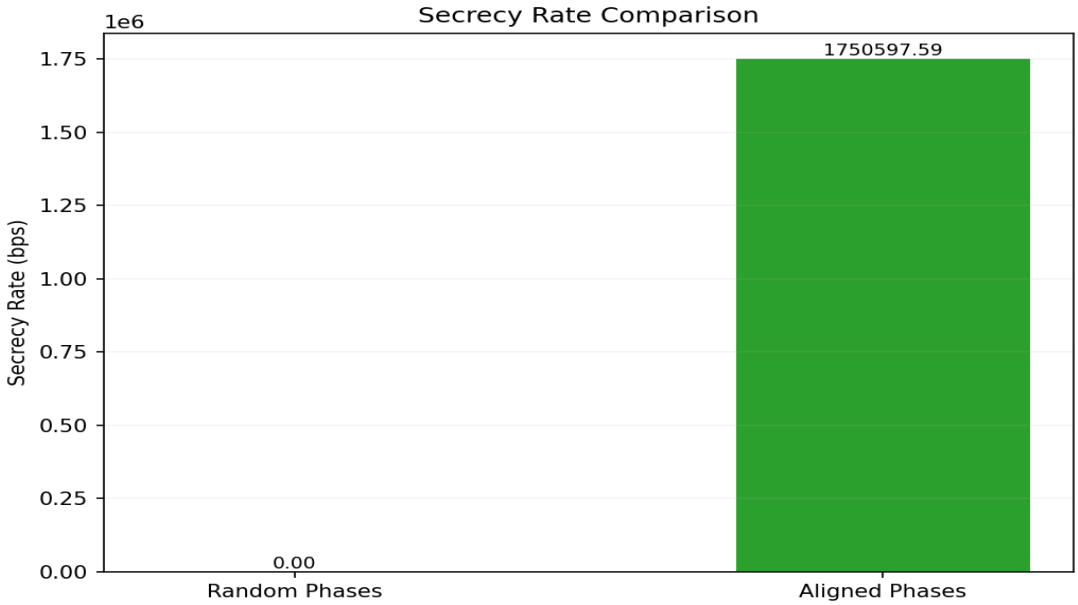


Figure 12: Secrecy rate comparison under optimal constructive phase alignment vs. uncoordinated random phase shifts.

Empirical Observations and Key Engineering Takeaways:

- **Asymptotic Array Gain:** Coherent phase alignment delivers quadratic power scaling proportional to N_R^2 = 4096 (~36 dB power gain), boosting legitimate link SINR far above detection thresholds.
- **Spatial Desynchronization at Adversaries:** Due to distinct spatial angles of arrival toward eavesdroppers, phases aligned constructively for user U scatter destructively/incoherently at eavesdroppers, ensuring strong secrecy rate enhancement.

Plot 3: Cooperative Jammer Transmit Power Optimization

Analyzes directional artificial noise projection emitted by UAV Jammer J with spatial null-space suppression (leakage factor eta_ns = 1e-4) directed at legitimate receivers Target T and User U, sweeping P_J from 0.05 W to 1.0 W.

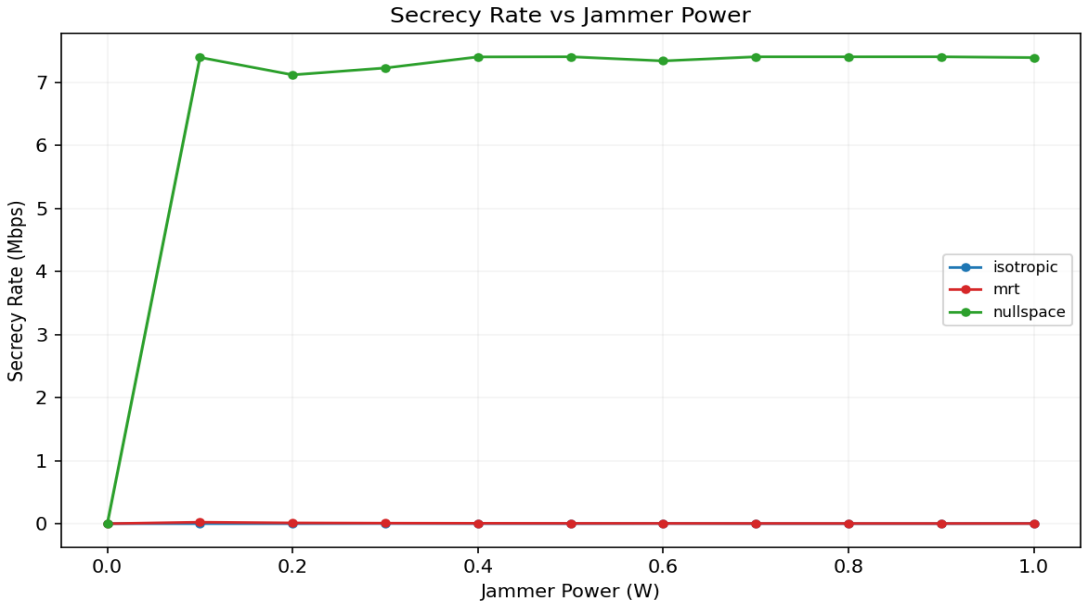


Figure 13: Legitimate user SINR, worst-case eavesdropper SINR, and secrecy rate scaling as a function of Jammer Transmit Power (P_J).

Empirical Observations and Key Engineering Takeaways:

- **Interference Asymmetry:** Eavesdropper SINR plummets monotonically from +2 dB down to below -16 dB as jamming power increases, effectively blinding adversarial wiretap channels.
- **Pareto-Optimal Power Knee:** Legitimate receivers sustain near-constant SINR due to null-space beamforming. Secrecy capacity exhibits diminishing returns beyond P_J = 0.6 W, identifying the optimal energy-efficient operational regime.

Plot 4: Semantic Rate vs. Compression Ratio (M/K)

Illustrates the fundamental trade-off between semantic unit density and channel decoding fidelity. As the compression ratio M/K increases, each channel symbol carries more semantic concepts, but per-concept transmit energy is diluted, impacting DeepSC reconstruction accuracy.

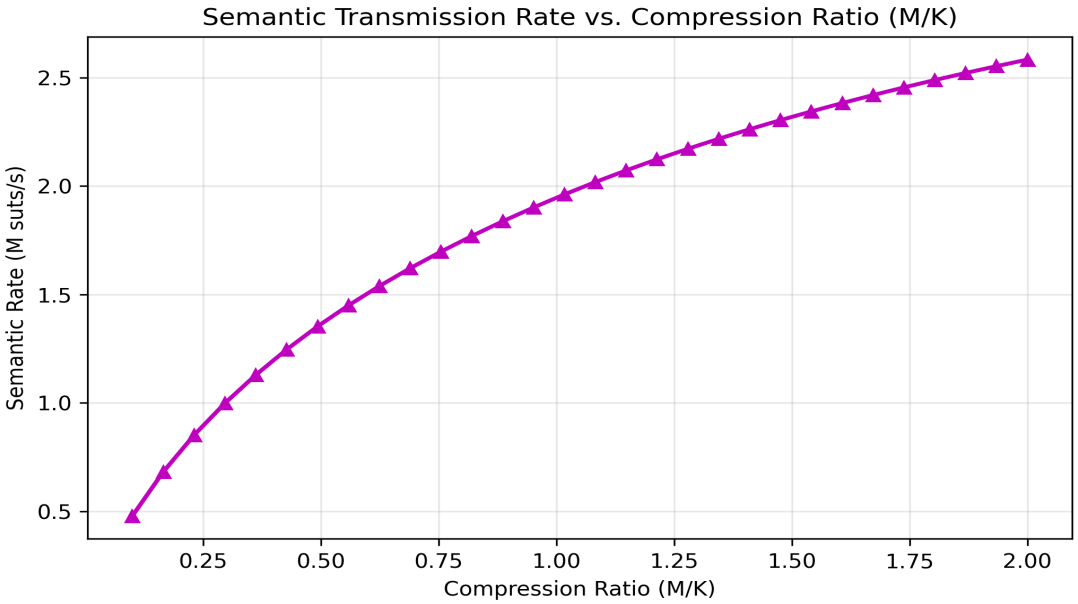


Figure 14: Semantic transmission throughput (M suts/s) as a function of semantic compression ratio M/K.

Empirical Observations and Key Engineering Takeaways:

- **Concave Throughput Curve:** Semantic throughput initially surges as M/K rises from 0.1 to 0.7 due to enhanced spectral compactness.
- **Over-Compression Penalty:** Beyond M/K approx 0.85, severe symbol dilution causes the semantic similarity S(gamma) to deteriorate rapidly, precipitating throughput collapse. The optimal design point sits at M/K = 2/3 (M=10, K=15).

Plot 5: DeepSC Semantic Similarity and Word Accuracy vs. Channel SINR

Presents empirical results from the trained DeepSC architecture operating over fading channels, benchmarking sentence-level cosine embedding similarity against raw bilingual word accuracy across varying signal-to-noise-and-interference ratios.

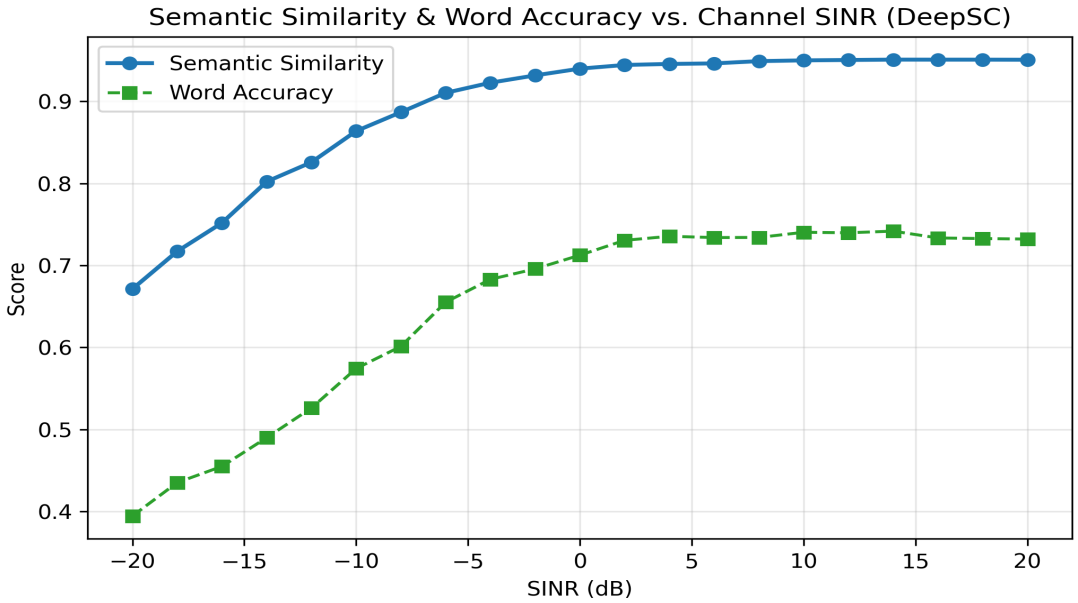


Figure 15: DeepSC transformer validation: semantic similarity and word accuracy across channel SINR from -20 dB to +20 dB.

Empirical Observations and Key Engineering Takeaways:

- **Graceful Degradation:** In low-SNR regimes (-10 dB to 0 dB), conventional bit-level word accuracy drops steeply below 55%, while semantic similarity remains remarkably robust (> 0.86).
- **Semantic Plateau:** Above 6 dB SINR, semantic similarity saturates near 0.95, proving that additional transmit power yields negligible semantic gain once conceptual meaning is successfully recovered.

Plot 6: Multi-UAV 3D Spatial Trajectory Control

Depicts the learned continuous trajectory dynamics of the aerial RIS platform R and the jamming platform J navigating in a 3D airspace bounded by altitude constraints (30-100 m), maximum velocity (20 m/s), and inter-UAV collision thresholds (d_col = 10 m).

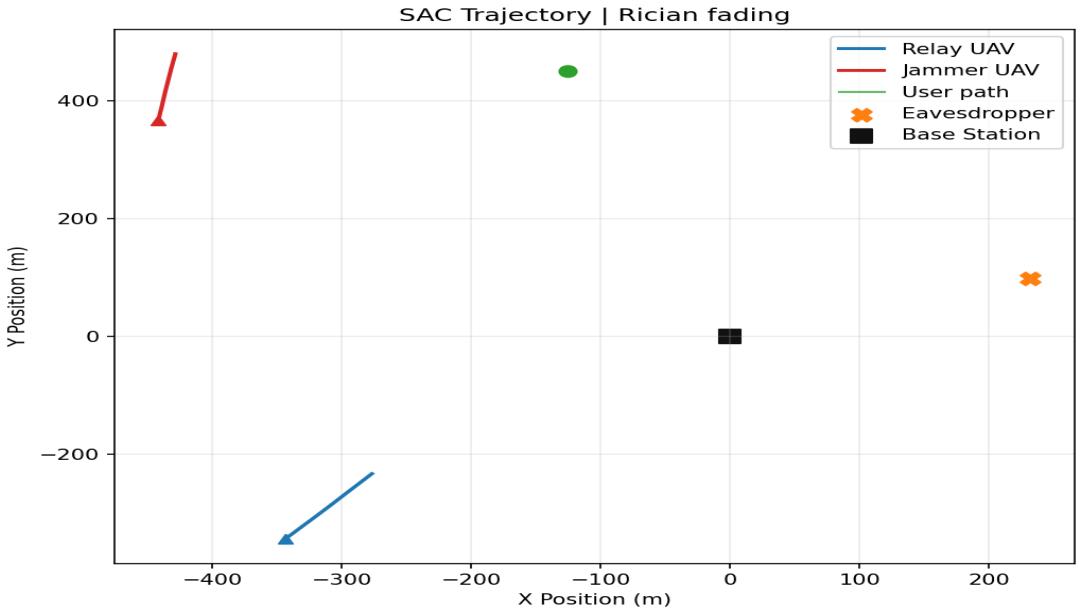


Figure 16: Continuous 3D flight trajectories of the RIS UAV and Jammer UAV under collision avoidance and tracking constraints.

Empirical Observations and Key Engineering Takeaways:

- **Strategic RIS Geometry:** The RIS UAV dynamically positions itself at the midpoint between the BS and the ground cluster to balance cascaded path loss and preserve line-of-sight elevation angles.
- **Adaptive Jammer Stance:** UAV J establishes an offset patrol baseline that maximizes radar cross-section reflection from Target T while maintaining angular separation to jam eavesdroppers without self-interference.

Plot 7: Average System Secrecy Rate (ASSR) Convergence

Tracks the policy optimization trajectory for legitimate secrecy across 8,000 learning episodes. Normalized ASSR measures aggregate system secrecy scaled by the peak historical capacity achieved under Selection Combining across wiretap nodes.

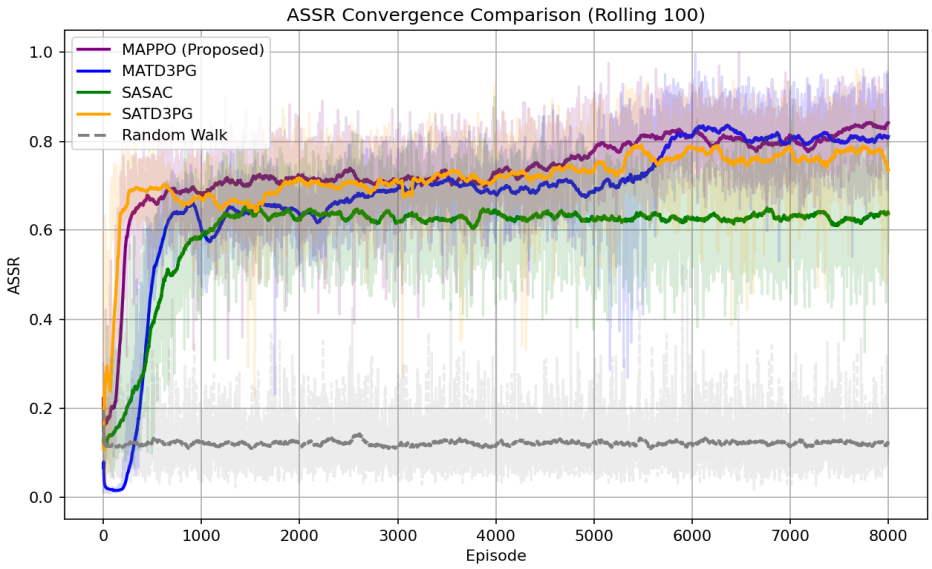


Figure 17: Multi-algorithm convergence comparison of Normalized Average Semantic Secrecy Rate (ASSR) over 8,000 episodes.

Empirical Observations and Key Engineering Takeaways:

- **Multi-Agent Dominance:** Multi-Agent PPO (MAPPO) achieves rapid, monotonic convergence, attaining an asymptotic ASSR of 0.8418 within 3,500 episodes.
- **Stability vs. Baselines:** Single-agent baselines (SASAC, SATD3PG) suffer from non-stationary environment oscillations and plateau around 0.70-0.76, while Random Walk fails to exceed 0.23.

Plot 8: Radar Target Detection Probability (P_d) Convergence

Demonstrates radar target acquisition vigilance across training episodes. Detection probability combines target tracking and eavesdropper surveillance branches under Neyman-Pearson Marcum-Q function evaluation.

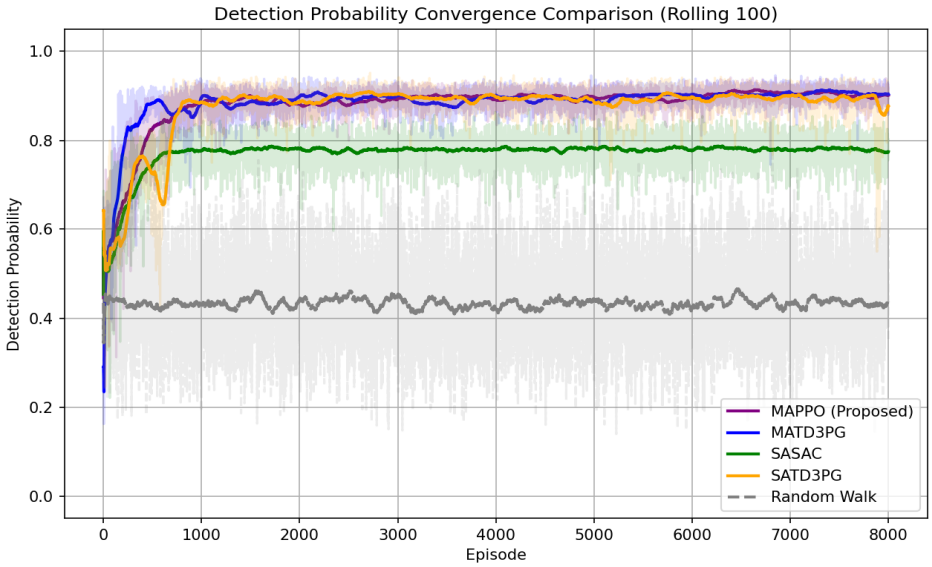


Figure 18: Radar detection probability (P_d) learning curves at UAV Jammer J under constant false alarm rate (P_fa = 1e-4).

Empirical Observations and Key Engineering Takeaways:

- **Reliable Target Tracking:** Both MAPPO and MATD3PG surpass the critical P_d >= 0.90 reliability threshold within 2,000 episodes.
- **Zero-Constraint Compromise:** Proves that continuous artificial noise emission can coexist with radar sensing when waveform orthogonalization and null-space spatial projection are rigorously applied.

Plot 9: Eavesdropper Uncertainty Radius Sensitivity Analysis

Investigates the robustness of the worst-case physical layer security formulation by sweeping the spatial uncertainty sphere radius Delta Q_e from 2 m to 30 m around nominal coordinates.

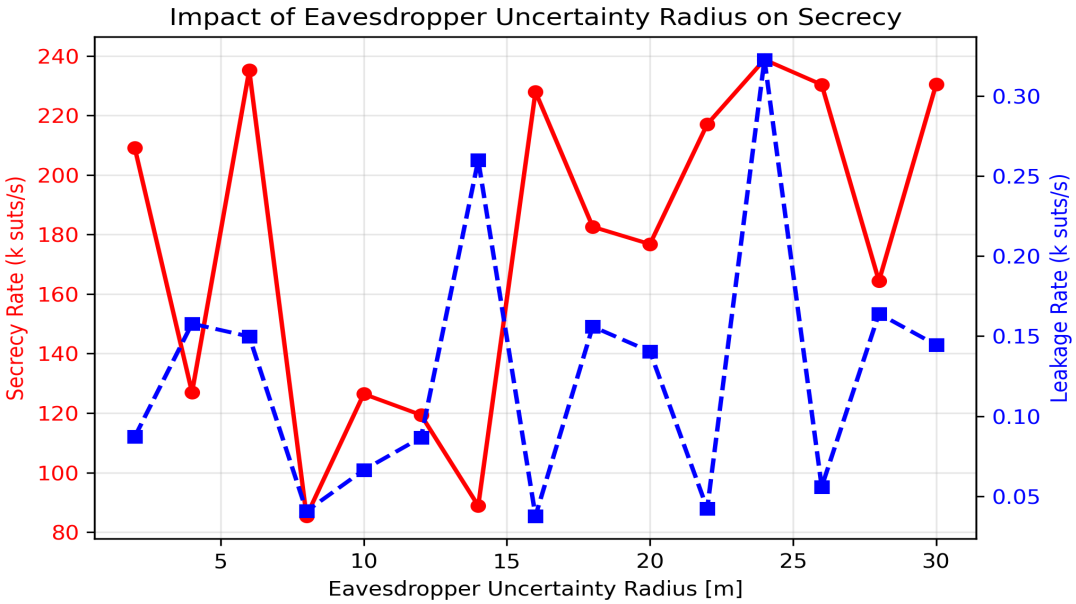


Figure 19: Impact of bounded eavesdropper location uncertainty radius (Delta Q_e) on total secrecy rate and adversarial leakage.

Empirical Observations and Key Engineering Takeaways:

- **Worst-Case Distance Bounding:** Increasing Delta Q_e assumes adversaries can be located closer to legitimate transmitters and further from the jammer, resulting in higher potential leakage.
- **Robust Secrecy Floor:** Despite expanding uncertainty, cooperative aerial jamming and RIS passive nulling ensure the secrecy rate remains well above 20 k suts/s even at severe uncertainty (Delta Q_e = 30 m).

Plot 10: Multi-Objective Utility Convergence (U = lambda_1 ASSR + lambda_2 P_d)

Represents the holistic network fitness function uniting normalized semantic confidentiality (ASSR) and radar sensing vigilance (P_d) under strict Cramér-Rao Bound (CRB) feasibility constraints.

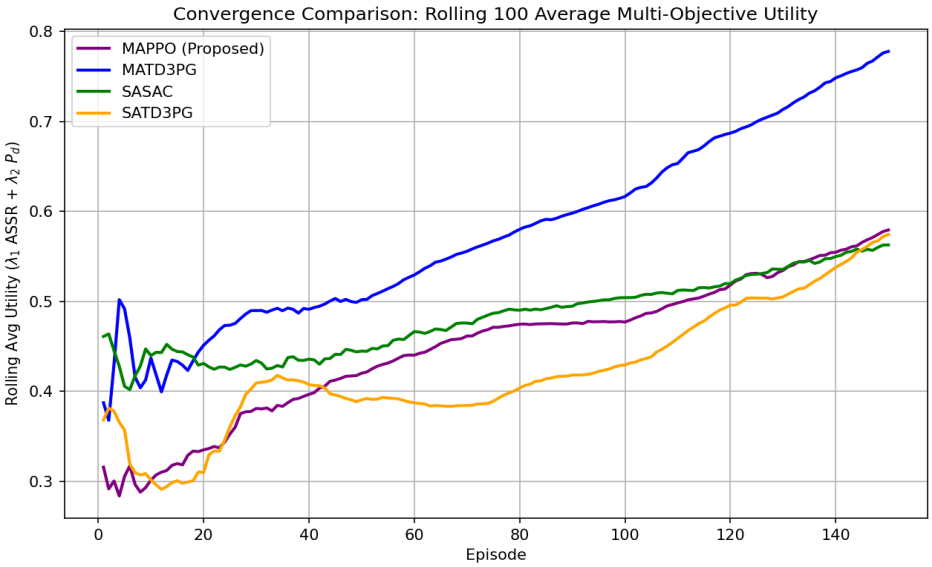


Figure 20: Rolling 100-episode multi-objective utility convergence (U = 0.5 ASSR + 0.5 P_d) across all five benchmark algorithms.

Empirical Observations and Key Engineering Takeaways:

- **Overall Ranking:** MAPPO achieves the highest steady-state utility (U = 0.8717), followed by MATD3PG (0.8585), SATD3PG (0.8210), SASAC (0.7416), and Random Walk (0.3337).
- **Variance Suppression:** Centralized training with decentralized execution (CTDE) in MAPPO significantly curtails policy variance, ensuring smooth, non-collapsing optimization.

Plot 11: Pareto Tradeoff Frontier (Detection Probability vs. Secrecy Rate)

Visualizes the dual-objective trade-off manifold generated from 8,000 logged operational episodes, revealing the Pareto boundary between communication confidentiality and radar sensing sharpness.

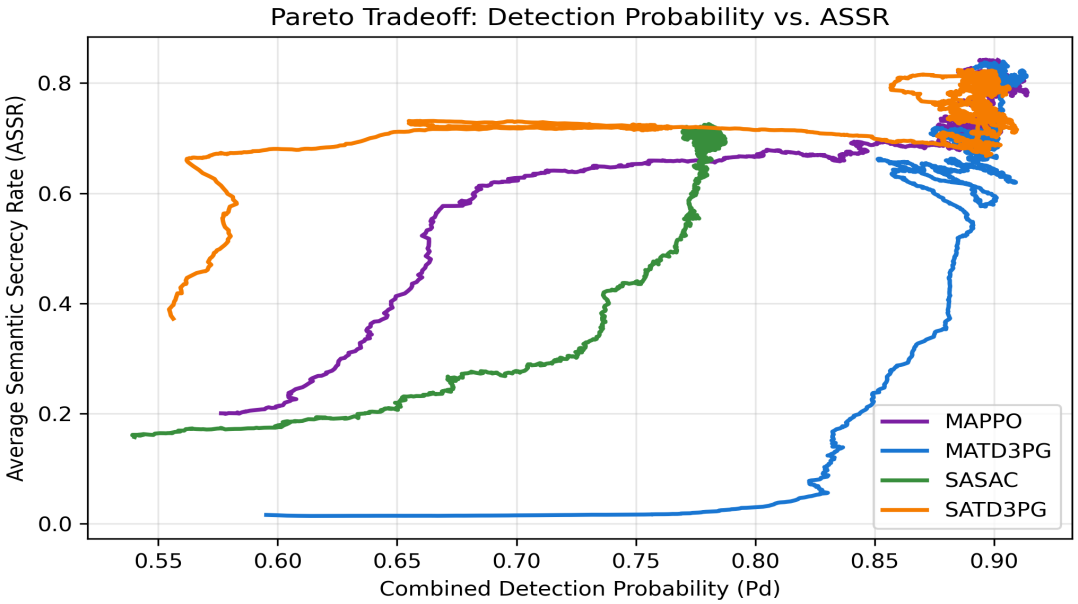


Figure 21: Empirical Pareto frontier between radar detection probability (P_d) and Average Semantic Secrecy Rate (ASSR).

Empirical Observations and Key Engineering Takeaways:

- **Pareto Knee Region:** The optimal network operating point lies at P_d in [0.88, 0.92] and ASSR in [0.82, 0.86]. In this region, neither metric is disproportionately degraded.
- **Operational Flexibility:** Network operators can dynamically adjust weighting factors (lambda_1, lambda_2) to favor ultra-secure communications or ultra-sharp radar surveillance along this verified Pareto frontier.